Special Topics in Operations Research

Michael Forbes and others

Description

This course will focus on teaching recent developments in Operations Research, with an emphasis on papers from the last five years. The following topics may be included:

- Fragment methods. These methods combine a priori column generation and Benders decomposition and have proven particularly successful for classes of vehicle routing problems.
- Stochastic optimisation - recent applications and techniques, including Stochastic Dual Dynamic Programming
- Mixed Integer Non-linear Programming (MINLP). The addition of a general purpose solver for MINLP to Gurobi 12 opens up a number of possibilities, which we will explore.
- Constraint Programming. For some optimisation problems (especially machine scheduling problems) constraint programming can be a useful solver in its own right, or it can be used to solve sub-problems in Benders decomposition or Branch and Price.
- Domain Independent Dynamic Programming. Again, this can be a useful solver in its own right, or for sub-problems.

Planned absence

- I will be away weeks 8 to 10 – all input before then
- Available for consultation by email
- Very limited ability to diagnose code issues, as I will not be able to run it

Learning objectives

- Apply and evaluate state of the art techniques suitable for solving large-scale problems in operations research
- Effectively communicate solutions to operations research problems in the style of a paper.

Assessment

The assessment will be based on an individual project in the style of MATH3205/7202 projects. Students, in consultation with the course co-ordinator, will select a recent paper (2021 or later) and look to replicate/extend the work in the paper. The specific assessment items will be:

- Initial presentation (20%): This will be an in class presentation, summarising the content of the paper chosen and the techniques which will be investigated. Students will be expected to respond to questions.
- Second presentation (20%): This will be an in class presentation, summarising the techniques investigated and the results achieved. Students will be expected to respond to questions.
- Project report (50%): This is the final report, including code, for the project. The final report will be formatted as a paper using the European Journal of Operations Research template.
- Participation (10%): Students will be assessed on the extent to which they contribute to online and classroom discussions of the materials.

Initial Presentation

- Your initial presentation is worth 20% of your assessment. It can be presented on either the 25th March or 1st April entirely at your discretion. Please email me at least one day in advance of your presentation to let me know you wish to present.
- The presentation should go for 10 to 15 minutes. You will be marked on the following criteria (5 marks available for each category).
- Presentation technique - How clear and engaging were your presentation and slides?
- Level and correctness of mathematics - Was the level of mathematics appropriate to the audience and was it correct? Were questions handled well?
- Summary of source paper - Was the source paper summarised so we could understand what it did and what the limitations and issues were, if any?
- Summary of your plan of attack - Did you describe how you are planning to proceed, including the collecting of data, initial replication and extensions ideas?

Final presentation

- Week 11 to 13
- 20 marks
- Presentation Technique – 5 marks
- Level and correctness of mathematics – 5 marks
- Summary of methods applied and results so far – 5 marks
- Lessons learned/insights/further plans – 5 marks

Project report

- Monday 8th June – 1pm
- Format of EJOR paper – 10 to 20 pages long
- Code and data required to run code
- 50 marks
- Appropriate academic writing – 10 marks
- Level and correctness of mathematics – 10 marks
- Description of methods applied and results – 10 marks
- Lessons learned/insights/possible future work – 10 marks
- Code correct and understandable – 10 marks